

1. Introduction

1.1 Purpose

This document specifies the software requirements for the **E-commerce Sports Shop System**. It provides a detailed description of system functionalities and constraints to support the development and testing process.

1.2 Scope

The system enables customers to browse sports products and place orders online. Admins can manage products, customers, and orders through a centralized system.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
Admin	User who manages system operations
Customer	User who purchases products
SRS	Software Requirements Specification
BR	Business Requirement
FR	Functional Requirement
NFR	Non-Functional Requirement

2. Overall Description

2.1 Product Perspective

The E-commerce Sports Shop System is a standalone web-based application designed to support online sales for a small sports shop.

2.2 User Classes and Characteristics

User	Description
Customer	Basic computer and phone skills, internet access
Admin	Trained staff managing the system

2.3 Operating Environment

- Web browser (Chrome, Edge, Firefox, safari, coc coc)
- Internet connection

2.4 Constraints

- No online payment integration in the current scope
- Designed for small-scale business operations

3. System Actors

Actor	Description
Customer	Browse products and place orders
Admin	Manage products and orders

System	E-commerce platform
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4. Functional Requirements

4.1 Customer Functional Requirements

- **FR-01:** The system shall allow customers to register an account.
- **FR-02:** The system shall allow customers to log in and log out.
- **FR-03:** The system shall allow customers to browse products by category.
- **FR-04:** The system shall display product details including name, price, and description.
- **FR-05:** The system shall allow customers to add products to a shopping cart.
- **FR-06:** The system shall allow customers to place an order.
- **FR-07:** The system shall allow customers to view order status.

4.2 Admin Functional Requirements

- **FR-08:** The system shall allow admin to create, update, and delete product information.
- **FR-09:** The system shall allow admin to view customer orders.
- **FR-10:** The system shall allow admin to update order status.
- **FR-11:** The system shall allow admin to manage customer accounts.

5. Non-Functional Requirements

5.1 Performance Requirements

- **NFR-01:** The system shall respond to user requests within 3 seconds under normal load.
- **NFR-02:** The system shall support at least 100 concurrent users.

5.2 Security Requirements

- **NFR-03:** User authentication shall be required to place orders.
- **NFR-04:** Admin functions shall be accessible only to authorized users.

5.3 Usability Requirements

- **NFR-05:** The system shall have a simple and user-friendly interface.
- **NFR-06:** The system shall be accessible via common web browsers.

6. Use Case Specification: Place Order

1. **Use Case Name:** Place Order

2. **Primary Actor:** Customer

3. **Brief Description:** Describes the process of a customer finalizing a purchase for items in their shopping cart after being authenticated by the system.

4. **Relationships:**

- **Includes:** [Login](#), [View Shopping Cart](#) (These steps are mandatory to complete the process).
- **Extends:** [Apply Coupon](#) (Optional – triggered only if the customer has a discount code).

5. **Preconditions:**

- The Customer has a valid account registered in the system.
- The Shopping Cart contains at least one product.

6. **Main Flow:**

Step	Actor Action	System Response
1	The Customer clicks the "Checkout" button from the shopping cart.	
2		Include(Login): The system checks the login status. If not logged in, it prompts the Customer to log in.

3		Include(View Shopping Cart): The system displays the order summary, including subtotal, shipping fees, and taxes.
4	The Customer confirms shipping information and selects a payment method.	The system validates the provided information.
5	The Customer clicks "Confirm Order".	The system records the order in the database and updates product inventory.
6		The system displays an order confirmation message and a unique Order ID.

7. Alternative Flows:

- **AF1. Empty Shipping Information:** At Step 4, if the customer leaves required fields (e.g., address, phone) blank, the system displays an error message and prevents order submission.
- **AF2. Out of Stock:** At Step 5, if an item in the cart becomes unavailable before the order is finalized, the system notifies the customer and updates the cart.
- **AF3. Order Cancellation:** The Customer can exit the checkout process at any time before Step 5; the system preserves the current cart state.

8. Postconditions:

- The order is successfully created in the system.
- The initial order status is set to "New".

7. Traceability (BR → FR Mapping)

Business Requirement	Functional Requirement
BR-01	FR-03

BR-03	FR-01, FR-02
BR-05	FR-06
BR-08	FR-08, FR-09

8. Assumptions and Dependencies

- Users have stable internet connections.
- Admin users are trained before system usage.